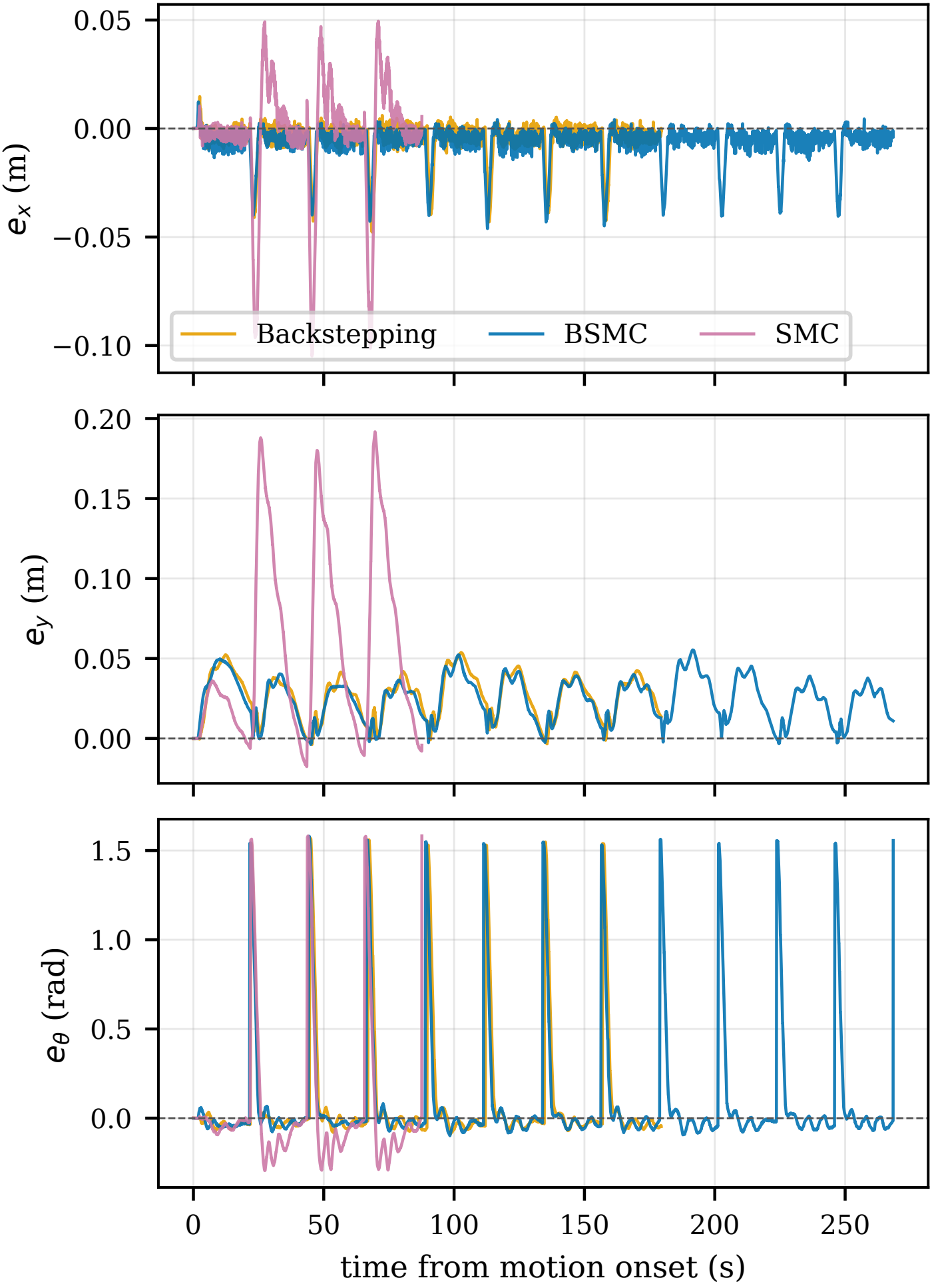

0.05
0.00
−0.05
−0.10
e_x (m)
Backstepping
BSMC
SMC
0.20
0.15
0.10
0.05
0.00
e_y (m)
1.5
1.0
0.5
0.0
e_θ (rad)
0
50
100
150
200
250
time from motion onset (s)